Elasticity of Demand

Elasticity is the ratio of percentage change in dependent variable to percentage change in any one of independent variables. It is a tool for measuring responsiveness of dependent variable to changes in an independent variable. Frequently used concepts of demand elasticity include price elasticity of demand; income elasticity of demand, and cross elasticity of demand.

Price Elasticity of Demand

Price elasticity of demand indicates the degree of responsiveness of quantity demanded of a good to change in its price, when other influencing factors such as income, prices of related goods that determine demand are held constant. Price elasticity of demand is the ratio of percentage change in quantity demanded of a commodity and percentage change in its price. Symbolically,

$$E_P = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$= \frac{\text{Change in quantity demanded}}{\text{Initial demand}} \div \frac{\text{Change in price}}{\text{Initial price}}$$

$$= \frac{\Delta Q}{Q} \div \frac{\Delta P}{P} = \frac{\Delta Q}{Q} \text{ x } \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \text{ x } \frac{P}{Q}$$

Where, E_P = Price elasticity of demand; ΔQ = Change in quantity demanded; ΔP = Change in price; Q = Initial demand; and, P = Initial price.

Income Elasticity of Demand

Income elasticity of demand indicates degree of responsiveness of quantity demanded of a good to change in consumer's income, when other factors such as price of the good under concern, prices of related goods, etc. that determine demand are held constant. Income elasticity of demand is the ratio of percentage change in quantity demanded of a commodity and percentage change in consumer's income. Symbolically,

$$E_Y = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

$$= \frac{\text{Change in quantity demanded}}{\text{Initial demand}} \div \frac{\text{Change in income}}{\text{Initial income}}$$

$$= \frac{\Delta Q}{Q} \div \frac{\Delta Y}{Y} = \frac{\Delta Q}{Q} \text{ x } \frac{Y}{\Delta Y} = \frac{\Delta Q}{\Delta Y} \text{ x } \frac{Y}{Q}$$

Where, E_Y = Income elasticity of demand; ΔQ = Change in quantity demanded;

ΔY = Change in income; Q = Initial demand; and, Y = Initial income.

Cross Elasticity of Demand

Cross elasticity of demand measures degree of responsiveness of quantity demanded of a good to change in price of a related good, either substitute or complementary, when other factors such as price of the good under concern, consumer's income, etc. that determine demand are held constant. Symbolically,

$$E_C = \frac{\text{Percentage change in quantity demanded of good X}}{\text{Percentage change in price of good Y}}$$

$$= \frac{\text{Change in quantity demanded of good X}}{\text{Initial demand for good X}} \div \frac{\text{Change in price of good Y}}{\text{Initial price of good Y}}$$

$$= \frac{\Delta Q_x}{Q_x} \div \frac{\Delta P_y}{P_y} = \frac{\Delta Q_x}{Q_x} \text{ x } \frac{P_y}{\Delta P_y} = \frac{\Delta Q_x}{\Delta P_y} \text{ x } \frac{P_y}{Q_x}$$

Where, E_C = Cross elasticity of demand; ΔQ_x = Change in quantity demanded of good X; ΔP_Y = Change in price of good Y; Q_x = Initial demand for good X; and, P_Y = Initial price of good Y.

Different Values of Price Elasticity of Demand

As per demand law, there is inverse relation between price and quantity demanded. Accordingly, estimated value of price elasticity of demand is negative. On the basis of absolute value of price elasticity of demand, there are five categories. These are: elasticity greater than one; unit elasticity; elasticity less than one; zero elasticity; and, infinite elasticity.

We have seen that,

$$E_P = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

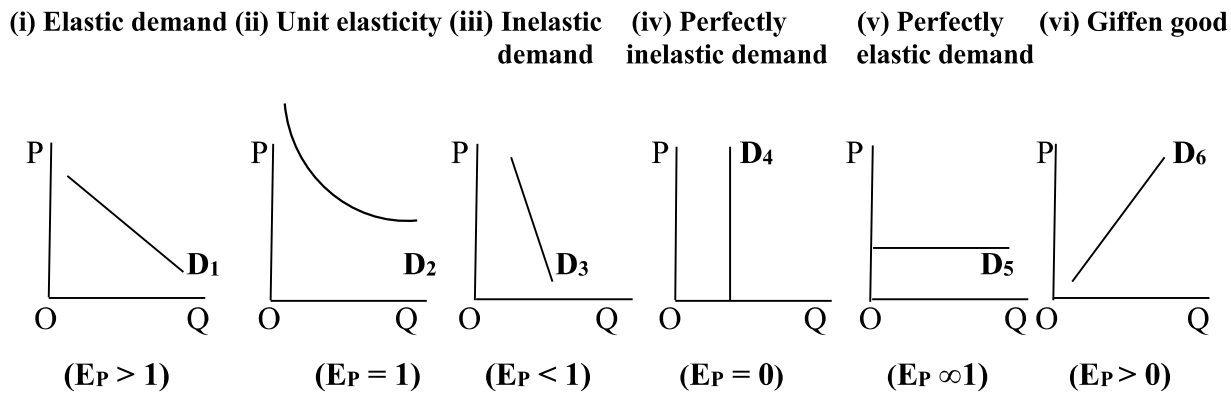
Hence, if due to fall in price, quantity demanded of good increases in a way that percentage increase in quantity is greater than percentage change in price; price elasticity of demand will be greater than unity. This type of demand is called elastic demand. Example of this type of commodities includes luxury items like, car, mansion, expensive trips, etc. If, due to fall in price, quantity demanded increases in a way that percentage increase in quantity demanded is equal to percentage change in price; price elasticity of demand will be equal to unity. When due to fall in price, quantity demanded of good increases in a way that percentage increase in quantity demanded is smaller than percentage change in price; price elasticity of demand will be less than

unity. This type of demand is called inelastic demand. Example of this type of commodities includes essential items like, food, clothing, medication, etc.

If for any reason, quantity demanded does not change due to change in price; it implies that price has no effect on quantity demanded. Consequently, price elasticity of demand becomes zero. This type of demand is called perfectly inelastic demand; perfect example of which is salt. In some cases, slight change in price of a good may lead to huge change in demand. Infinite elasticity is the case when large change in quantity demanded occurs due to insignificant change in price. This type of demand is called perfectly elastic demand. This type of demand is faced by a perfectly competitive firm.

As depicted by demand law, in general, there is an inverse relationship between price and quantity demanded. Where this law holds, goods are called normal goods. If, contrarily, there exists direct relation between quantity demanded of a good and its price, then that good is called Giffen good according to economist Sir Robert Giffen. A Giffen good is a low income, non-luxury product that defies standard economic and consumer demand theory. Demand for Giffen goods rises when the price rises and falls when the price falls. Examples of Giffen goods include low qualities of bread, rice, wheat, etc.; whose price elasticity of demand will be positive.

Figure 4: Different Values of Price Elasticity of Demand



Value of Income Elasticity of Demand and its Implication

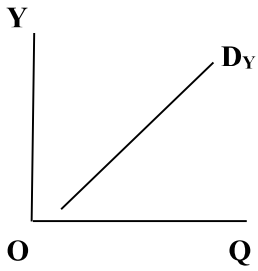
We have seen that, $$E_Y = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

As like price elasticity of demand, income elasticity of demand can also be classified into five categories like, elasticity greater than one; unit elasticity; elasticity less than one; zero elasticity; and, infinite elasticity. When due to change in income, quantity demanded of good changes in a way that percentage change in quantity demanded is greater than percentage change in income; income elasticity of demand will be greater than unity. This type of demand is called elastic demand. Example of this type of commodities includes luxury items like, car, mansion, expensive trips, etc. When due to change in income, quantity demanded changes in a way that percentage change in quantity demanded is equal to percentage change in income; income elasticity of demand is unity.

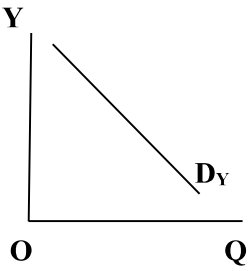
When due to change in income, quantity demanded of good changes in a way that percentage change in quantity demanded is smaller than percentage change in income; income elasticity of demand will be less than unity. This type of demand is called inelastic demand. Example of this type of commodities includes essential items like, food, clothing, medication, etc. If for any reason, quantity demanded does not change due to change in income; it implies that income has no effect on quantity demanded. Consequently, income elasticity of demand becomes zero. This type of demand is called perfectly inelastic demand. In cases, if slight change in income leads to huge change in demand; then, it is a case of infinite elasticity.

In case of E_Y, instead of absolute value, concern is more with respect to sign associated with this value. Accordingly, focus is on whether $E_Y > 0$ or $E_Y < 0$. In former case, income demand curve is upward sloping; implying that there is positive relation between income and demand. In latter case, income demand curve is downward sloping; implying that there is negative relation between income and demand. If, $E_Y > 0$, goods are called normal goods; and, if, $E_Y < 0$, goods are called inferior goods. Example of normal goods is generally consumed items of goods and inferior goods include coarse rice, cloth, etc.

Figure 7: Type of Good on the basis of Slope of Income Demand Curve



(a) Normal good ($E_Y > 0$)



(b) Inferior good ($E_Y < 0$)

Value of Cross Elasticity of Demand and its Implication

We have seen that,

$$E_C = \frac{\text{Percentage change in quantity demanded of good X}}{\text{Percentage change in price of good Y}}$$

Like E_Y, in case of E_C also, instead of absolute value, concern is more with respect to sign associated with this value. Accordingly, focus is on whether $E_C > 0$ or $E_C < 0$ or $E_C = 0$. As the above formulation of E_c suggests; if $E_C > 0$, it implies that with increase in price of good Y, there is increase in demand for good X. That is good X and Y is substitute of each other.

Following same reasoning, if $E_C < 0$; it implies that with increase in price of good Y, there is decrease in demand for good X. That is good X and Y is complementary to each other. When, $E_C = 0$; it implies that with change in price of good Y, there is no change in demand for good X. That is good X and Y is unrelated to each other.

Price Elasticity of Supply

Price elasticity of supply (E_S) is the measurement of responsiveness of quantity supplied of a commodity to change in its price. Price elasticity of supply is the ratio of percentage change in quantity supplied of a commodity and percentage change in its price. Symbolically,

$$E_S = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

$$= \frac{\text{Change in quantity supplied}}{\text{Initial supply}} \div \frac{\text{Change in price}}{\text{Initial price}}$$

$$= \frac{\Delta Q}{Q} \div \frac{\Delta P}{P} = \frac{\Delta Q}{Q} \text{ X } \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \text{ X } \frac{P}{Q}$$

Where, E_S = Price elasticity of supply; ΔQ = Change in quantity supplied;

ΔP = Change in price; Q = Initial supply; and, P = Initial price.

Different Values of Price Elasticity of Supply

As per supply law, there is direct relation between price and quantity supplied. Accordingly, estimated value of price elasticity of supply is positive. On the basis of absolute value of price elasticity of supply, there are five categories. These are: elasticity greater than one; unit elasticity; elasticity less than one; zero elasticity; and, infinite elasticity.

We have seen that,

$$E_S = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

Hence, if due to change in price, quantity supplied of good changes in a way that percentage increase in quantity is greater than percentage change in price; price elasticity of supply will be greater than unity. This type of supply is called elastic supply. If, due to change in price, quantity supplied changes in a way that percentage change in quantity supplied is equal to percentage change in price; price elasticity of supply will be equal to unity.

When due to change in price, quantity supplied of good changes in a way that percentage change in quantity supplied is smaller than percentage change in price; price elasticity of supply will be less than unity. This type of supply is called inelastic supply. If for any reason, quantity supplied does not change due to change in price; it implies that price has no effect on quantity supplied. Consequently, price elasticity of supply becomes zero. This type of supply is called perfectly inelastic supply; example of which is supply of perishable goods. In some cases, slight change in price of a good may lead to huge change in supply. Infinite elasticity is the case when large change in quantity supplied occurs due to insignificant change in price. This type of supply is called perfectly elastic supply.